EXERCISE 3: SQL CASE STATEMENTS

1. Table: products

product_id	Product_name	price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

```
SELECT product_name,
       price,
CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
    WHEN price < 100 THEN 'Budget'
END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2. Table: orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

```
SELECT customer_name,
       amount,
CASE
    WHEN amount >= 1000 THEN 'High Value'
    WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
    WHEN amount < 500 THEN 'Low Value'
END AS order_value_category
FROM orders;
```

customer_name	amount	order_value_category
Alice	150.00	Expensive
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	120.00	High Value

3. Table : employees

emp_id	emp_name	department	salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

```
SELECT emp_name,
       department,
       salary,

CASE
    WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
    ELSE 'Staff'
END AS position_level
FROM employees;
```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Table : students

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

```
SELECT student_name,
       score,
       ~~grade~~
CASE
    WHEN score >= 90 THEN 'A'
    WHEN score BETWEEN 80 AND 89 THEN 'B'
    WHEN score BETWEEN 70 AND 79 THEN 'C'
    WHEN score BETWEEN 60 AND 69 THEN 'D'
    WHEN score < 60 THEN 'F'
END AS grade
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	D
David	83	B
Ella	68	D

5. Table: deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

```
SELECT delivery_id
       delivery_time_minutes,
CASE
    WHEN delivery_time_minutes <= 30 THEN 'Fast'
    WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
    WHEN delivery_time_minutes > 60 THEN 'Late'
END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. Table: tickets

ticked_id	issue_type	priority
1	Long issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

```
SELECT issue_type,
       priority,

CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'

END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Long issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Table : attendance

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

```
SELECT student_id,
       attendance_percentage,
CASE
    WHEN attendance_percentage >= 90% THEN 'Excellent'
    WHEN attendance_percentage BETWEEN 75% AND 89% THEN 'Good'
    WHEN attendance_percentage < 75% THEN 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendane_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

8. Table : products_inventory

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

```
SELECT product_id,
       stock_qty,
CASE
   WHEN stock_qty = 0 THEN 'Out of Stock'
   WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
   WHEN stock_qty > 5 THEN 'In Stock'
END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Q. Table : classes

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

```
SELECT subject,
       enrolled_students,

CASE
    WHEN enrolled_students >= 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    WHEN enrolled_students < 0 THEN 'Small'

END AS class_size_category
FROM classes;
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Table : payments

payment-id	amount	payment-method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash

```
SELECT payment-id,
       payment-method,
       amount,
CASE
    WHEN payment-method = 'Cash' AND amount >= 200 THEN 'Eligible
                                                         for
                                                         Discount'
    ELSE 'Not Eligible'
    END AS discount-eligibility
    FROM payments;
```

payment-id	payment-method	amount	discount-eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount